

数字信号处理课程教学大纲					
课程基本信息 (Course Information)					
课程代码 (Course Code)	ICE4405P	学时 (Credit Hours)	48	学分 (Credits)	3.0
课程名称 (Course Name)	(中文)数字信号处理				
	(英文)Digital Signal Processing				
课程性质 (Course Type)	专业必修课程				
授课对象 (Target Audience)	Third-Year Undergraduates Majoring in Information Engineering				
授课语言 (Language of Instruction)	全英文				
开课院系 (School)	巴黎卓越工程师学院				
先修课程 (Prerequisite)	大学基础物理	后续课程 (post)			
课程负责人 (Instructor)	李龙光	课程网址 (Course Webpage)			
课程简介 (中文) (Description)	本课程涵盖数字信号处理(DSP)的基本理论，并着重于DSP的实际应用。它包括四个部分:(1)信号与系统简介;(2)采样与傅里叶分析;(3)滤波器设计与分析;(4)随机信号处理与频谱分析。每个模块包括公式推导、数值计算、工程应用和研究报告。				
	在第一部分中，介绍信号与系统概念，给出了基本连续时间(CT)和离散时间(DT)信号，并利用脉冲响应函数分析了连续时间(CT)和离散时间(DT) LTI系统的特性。对CT系统和DT LTI系统的响应进行了卷积计算。				
	作为第二部分，CT和DT信号之间的联系由采样定理表示。详细说明了抽样定理所涉及的数学和物理意义。给出了CT信号傅里叶变换的实现方法，讨论了傅里叶变换的性质。然后比较分析了CT傅立叶变换与离散傅里叶变换(DTFT)、离散傅里叶变换(DTFT)和离散傅里叶变换(DFT)之间的联系，以及DFT的快速算法。引入傅里叶级数(FS)作为傅里叶级数的扩展，探讨了窗口增加带来的影响。并给出了一个与傅立叶变换相关的应用实例。				
	在第三部分中，系统的性质被讨论的频率响应函数和传递函数。介绍了典型的模拟滤波器。然后介绍了FIR和IIR滤波器的设计和分析。利用MATLAB提供的滤波器设计工具，构造符合规范的滤波器。作为一个实例，提供了与数字滤波器有关的应用。				
	最后一个部分是随机变量和平稳随机信号的信号处理的介绍。介绍了自协方差估计、功率谱能量估计和传递函数估计。以维纳滤波器为例，讨论了系统的识别过程和系统校正方法。				

课程简介 (英文) (Description)	This course covers fundamental theories of digital signal processing (DSP) and focuses on practical applications of DSP. There are four modules, which are (1)introduction of signals and systems, (2) sampling and Fourier analysis, (3) filter design and analysis, and (4) random signal processing and spectral analysis. Each module includes the formula derivation, numerical calculations, engineering applications and a study report. In the first module, the concept of signals and systems is introduced and the basic continuous-time (CT) and discrete-time (DT) signals are represented, and the properties of CT and DT LTI systems are analyzed by the impulse response function. The convolution calculations are carried out to calculate the responses of CT and DT LTI systems. As the second module, the link between CT and DT signals is formulated by the sampling theorem. The mathematical and physical meanings involved in the sampling theorem are explained in details. Realization of Fourier transform (FT) of a CT signal is demonstrated and the properties of FT are discussed. Then the connections between CT FT and the discrete-time Fourier transform (DTFT), DTFT and discrete Fourier transform (DFT), and the fast algorithm of DFT are investigated by comparisons. Fourier series (FS) is introduced as an extension of FT. Effects due to adding window are explored. An application related to FT is provided as a practical example. In the third module, the properties of systems are discussed in terms of frequency response functions and transfer functions. Typical analog filters are introduced. And then the design and analysis of FIR and IIR filters are introduced. The filter design tool provided by MATLAB is employed to construct filter with specifications. An application related to digital filters is provided as a practical example. The final module is the introduction of random variables and the signal processing for stationary random signals. Auto covariance, power spectral energy and transfer function estimation are introduced. The case of the Wiener filter is studied to show the system identification process and the system rectification method.							
课程目标与内容 (Course objectives and contents)								
课程目标 (Course Object)	1.Understand the relationship between continuous signals and discrete signals; 2.Understand the relationship between time-domain signals and frequency-domain signals with Fourier transform; 3.Understand how to perform signal filtering; 4.Understand how to digitalize an analog signal and be able to compute basic features of digital signals.							
教学内容 进度安排及对应课程 目标 (Class Schedule & Requirements & Course Objectives)	章节	教学内容 (要点)	学时	教学形式	作业及考核要求	课程思政融入点	对应课程目标	
	1	Introduction of signals and systems	8	Theory and exercise	Homework and Lab Report	Beidou System	Understand the relationship between continuous signals and discrete signals	

	2	Fourier transform	14	Theory and exercise	Homework and Lab Report	Handle Beidou signal	Understand the relationship between time-domain signals and frequency-domain
	3	Digital filter design and filtering	18	Theory and exercise	Homework and Lab Report	Remove noise from Beidou signal	Understand how to perform signal filtering
教学内容 进度安排及对应课程 目标 (Class Schedule & Requirements & Course Objectives)	4	Random signal analysis	8	Theory and exercise	Homework	Analysis noise in Beidou signal	Understand how to digitalize an analog signal and can compute basic features of digital signal
考核方式 (Grading)	Attendance 10%, Homework and Quiz 30%, Lab Project 20% (3 projects), Final Exam 40%.						
	教材名称	作者	出版社	出版日期	版次	书号	
	Discrete-Time Signal Processing	A.V. Oppenheim and R.W. Schafer	Pearson	2009	3	0131988425	
	Digital Signal Processing: Principles, Algorithms and Applications	J.G. Proakis and D.G. Manolakis,	Prentice Hall	2007	4	7121204371	
	Understanding Digital Signal Processing	Richard G. Lyons	Prentice Hall	2011	3	0137027419	
其它(More)							
备注(Notes)							